

Answer the problems on **separate** paper. You do not need to rewrite the problem statements on your answer sheets. Do your own work. Show **all relevant steps** which lead to your solutions. Attach this question sheet to the front of your answer sheets.

1. (16 pts) Let  $P(-4,-2)$  and  $Q(3,2)$  be points in the plane.

- A. Plot  $P$  and  $Q$  on a Cartesian coordinate system
- B. Find the distance between  $P$  and  $Q$
- C. Find the coordinates of the midpoint  $M$  of the line segment  $\overline{PQ}$
- D. Find the equation of the line passing through  $P$  and  $Q$ . Write the equation in standard form.

2. (8 pts) Solve the equation  $|5 + 3y| = 4$ .

3. (8pts) Solve algebraically the system of equations  $\begin{cases} 3x - 8y = 2 \\ 2x + 4y = 6 \end{cases}$

4. (12 pts) Let  $f(x) = \frac{x}{\sqrt{1-2x}}$ .

- A. Find the domain of  $f$
- B. Compute the following functional values or state that the corresponding  $x$ -value is not in the domain of  $f$

B1.  $f(-2)$

B2.  $f(-1)$

B3.  $f(1)$

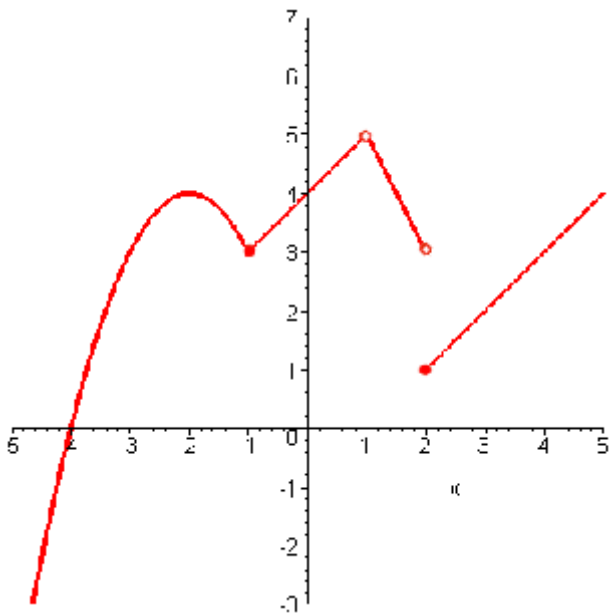
5. (12 pts) Let  $f(x) = 3 - 2x$ .

- A. Find the inverse of  $f$
- B. Sketch the graphs of  $f$  and  $f^{-1}$  on same coordinate system.

6. (6pts) Find the exact value of  $\tan(\cos^{-1} \frac{1}{2})$ .

7. (6pts) Simplify  $\sin(\cos^{-1} x)$

8. (12pts) Consider the function  $f$  defined by the graph to the right. Find each of the following (if they exist). If they do not exist, state so.



A1.  $f(-1)$

A2.  $f(1)$

A3.  $f(2)$

B1.  $\lim_{x \rightarrow -1^-} f(x)$

B2.  $\lim_{x \rightarrow 1^-} f(x)$

B3.  $\lim_{x \rightarrow 2^-} f(x)$

C1.  $\lim_{x \rightarrow -1^+} f(x)$

C2.  $\lim_{x \rightarrow 1^+} f(x)$

C3.  $\lim_{x \rightarrow 2^+} f(x)$

D1.  $\lim_{x \rightarrow -1} f(x)$

D2.  $\lim_{x \rightarrow 1} f(x)$

D3.  $\lim_{x \rightarrow 2} f(x)$

9. (24 pts) Algebraically evaluate each of the following limits.

A.  $\lim_{x \rightarrow 1} \frac{x^2 - 2x - 7}{x^2 + x - 1}$

B.  $\lim_{x \rightarrow -3} \frac{6 - x - x^2}{3 + x}$

C.  $\lim_{x \rightarrow 0} \frac{\sin^2 2x}{x^2}$